PROJECT NAME: VULNERABILITY MANAGEMENT DATABASE FOR OPEN-SOURCE SOFTWARE (VMDB-OSS) – EXECUTION IMAGES

BACHELOR OF BUSINESS ADMINISTRATION (Hons.) IN ARTIFICIAL INTELLIGENCE
VERSION – 1.1

Project Under: Database Management System, Mini Project Submission

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VMDB - OSS - Query

VMDB_OSS: Vulnerability Management Database for Open Source Software

VMDB_OSS: Schema & Query Report

Version: 1.1 **Author:** Ashwin Chhawaniya **Date:** November 2025

Introduction

This document provides a line-by-line breakdown of the `VMDB_OSS` database schema. For each database object (Table, View, Procedure, Function), it includes:

1. A brief description of its purpose.
2. A `DESCRIBE` or `SHOW CREATE` command to inspect its structure.
3. A placeholder for the screenshot of the structure's output.
4. A `SELECT *` or `CALL` command to query its data.
5. A placeholder for the screenshot of the query's output.

1.0 Tables

This section details the 7 core tables that form the foundation of the database.

1.1 `Users` Table

- **Purpose:** Stores user accounts and their roles (Admin, Maintainer, Analyst).

Structure Query:

```
DESCRIBE Users;
```

Structure Output:

```
mysql> DESC USERS;
+------------+--------------------------------------+------+-----+-------------------+-------------------+
| Field      | Type                                 | Null | Key | Default           | Extra             |
+------------+--------------------------------------+------+-----+-------------------+-------------------+
| user_id    | int                                  | NO   | PRI | NULL              | auto_increment    |
| full_name  | varchar(100)                         | NO   |     | NULL              |                   |
| email      | varchar(100)                         | NO   | UNI | NULL              |                   |
| role       | enum('Admin','Maintainer','Analyst') | YES  |     | Maintainer        |                   |
| created_at | timestamp                            | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------+--------------------------------------+------+-----+-------------------+-------------------+
5 rows in set (0.06 sec)
```

Data Query:

```
SELECT * FROM Users;
```

Data Output:

```
mysql> SELECT*FROM USERS;
+---------+---------------+------------------+------------+---------------------+
| user_id | full_name     | email            | role       | created_at          |
+---------+---------------+------------------+------------+---------------------+
|       1 | Ravi Kumar    | ravi@vmdb.com    | Admin      | 2025-11-13 20:03:43 |
|       2 | Neha Patel    | neha@vmdb.com    | Analyst    | 2025-11-13 20:03:43 |
|       3 | Amit Shah     | amit@vmdb.com    | Maintainer | 2025-11-13 20:03:43 |
|       4 | Sarah Lee     | sarah@vmdb.com   | Maintainer | 2025-11-13 20:03:43 |
|       5 | John Carter   | john@vmdb.com    | Analyst    | 2025-11-13 20:03:43 |
|       6 | Priya Mehta   | priya@vmdb.com   | Admin      | 2025-11-13 20:03:43 |
|       7 | Mark Thompson | mark@vmdb.com    | Maintainer | 2025-11-13 20:03:43 |
|       8 | Olivia Brown  | olivia@vmdb.com  | Analyst    | 2025-11-13 20:03:43 |
|       9 | Liam Wilson   | liam@vmdb.com    | Maintainer | 2025-11-13 20:03:43 |
|      10 | Sophia Davis  | sophia@vmdb.com  | Admin      | 2025-11-13 20:03:43 |
+---------+---------------+------------------+------------+---------------------+
10 rows in set (0.08 sec)
```

1.2 `Projects` Table

- **Purpose:** Stores the software projects being tracked and links them to a maintainer.

Structure Query:

```
DESCRIBE Projects;
```

Structure Output:

```
mysql> DESC PROJECTS;
+---------------+--------------+------+-----+-------------------+-------------------+
| Field         | Type         | Null | Key | Default           | Extra             |
+---------------+--------------+------+-----+-------------------+-------------------+
| project_id    | int          | NO   | PRI | NULL              | auto_increment    |
| project_name  | varchar(100) | NO   |     | NULL              |                   |
| repo_url      | varchar(255) | YES  |     | NULL              |                   |
| maintainer_id | int          | YES  | MUL | NULL              |                   |
| created_at    | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+---------------+--------------+------+-----+-------------------+-------------------+
5 rows in set (0.00 sec)
```

Data Query:

```
SELECT * FROM Projects;
```

Data Output:

```
mysql> SELECT * FROM Projects;
+------------+--------------+-------------------------------------------+---------------+---------------------+
| project_id | project_name | repo_url                                  | maintainer_id | created_at          |
+------------+--------------+-------------------------------------------+---------------+---------------------+
|          1 | Flask        | https://github.com/pallets/flask          |             3 | 2025-11-13 20:03:54 |
|          2 | React        | https://github.com/facebook/react         |             4 | 2025-11-13 20:03:54 |
|          3 | Django       | https://github.com/django/django          |             6 | 2025-11-13 20:03:54 |
|          4 | TensorFlow   | https://github.com/tensorflow/tensorflow  |             3 | 2025-11-13 20:03:54 |
|          5 | ExpressJS    | https://github.com/expressjs/express      |             7 | 2025-11-13 20:03:54 |
|          6 | NumPy        | https://github.com/numpy/numpy            |             9 | 2025-11-13 20:03:54 |
|          7 | Pandas       | https://github.com/pandas-dev/pandas      |             2 | 2025-11-13 20:03:54 |
|          8 | Angular      | https://github.com/angular/angular        |             4 | 2025-11-13 20:03:54 |
|          9 | VueJS        | https://github.com/vuejs/core             |             7 | 2025-11-13 20:03:54 |
|         10 | FastAPI      | https://github.com/tiangolo/fastapi       |             9 | 2025-11-13 20:03:54 |
+------------+--------------+-------------------------------------------+---------------+---------------------+
10 rows in set (0.04 sec)
```

1.3 `Packages` Table

- **Purpose:** Master list of all unique open-source packages (e.g., 'Werkzeug', 'Lodash').

Structure Query:

```
DESCRIBE Packages;
```

Structure Output:

```
mysql> DESCRIBE PACKAGES;
+----------------+---------------------------------------------------------+------+-----+-------------------+-------------------+
| Field          | Type                                                    | Null | Key | Default           | Extra             |
+----------------+---------------------------------------------------------+------+-----+-------------------+-------------------+
| package_id     | int                                                     | NO   | PRI | NULL              | auto_increment    |
| package_name   | varchar(100)                                            | NO   |     | NULL              |                   |
| ecosystem      | enum('npm','PyPI','Maven','Go','RubyGems','Other')      | YES  |     | Other             |                   |
| latest_version | varchar(50)                                             | YES  |     | NULL              |                   |
| created_at     | timestamp                                               | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+----------------+---------------------------------------------------------+------+-----+-------------------+-------------------+
5 rows in set (0.01 sec)
```

Data Query:

```
SELECT * FROM Packages;
```

Data Output:

```
mysql> SELECT*FROM PACKAGES;
+------------+--------------+-----------+----------------+---------------------+
| package_id | package_name | ecosystem | latest_version | created_at          |
+------------+--------------+-----------+----------------+---------------------+
|          1 | Werkzeug     | PyPI      | 3.0.1          | 2025-11-13 20:04:01 |
|          2 | Jinja2       | PyPI      | 3.1.4          | 2025-11-13 20:04:01 |
|          3 | React        | npm       | 18.3.0         | 2025-11-13 20:04:01 |
|          4 | Express      | npm       | 4.19.2         | 2025-11-13 20:04:01 |
|          5 | TensorFlow   | PyPI      | 2.16.1         | 2025-11-13 20:04:01 |
|          6 | NumPy        | PyPI      | 1.26.4         | 2025-11-13 20:04:01 |
|          7 | Pandas       | PyPI      | 2.2.0          | 2025-11-13 20:04:01 |
|          8 | Lodash       | npm       | 4.17.21        | 2025-11-13 20:04:01 |
|          9 | Django       | PyPI      | 5.1.1          | 2025-11-13 20:04:01 |
|         10 | FastAPI      | PyPI      | 0.110.0        | 2025-11-13 20:04:01 |
+------------+--------------+-----------+----------------+---------------------+
10 rows in set (0.03 sec)
```

1.4 `Vulnerabilities` Table

- **Purpose:** Master list of all known vulnerabilities (CVEs), each linked to a specific package.

Structure Query:

```
DESCRIBE Vulnerabilities;
```

Structure Output:

```
mysql> DESC VULNERABILITIES;
+----------------+-------------------------------------------+------+-----+-------------------+-------------------+
| Field          | Type                                      | Null | Key | Default           | Extra             |
+----------------+-------------------------------------------+------+-----+-------------------+-------------------+
| vuln_id        | int                                       | NO   | PRI | NULL              | auto_increment    |
| cve_id         | varchar(50)                               | YES  | UNI | NULL              |                   |
| package_id     | int                                       | YES  | MUL | NULL              |                   |
| summary        | text                                      | YES  |     | NULL              |                   |
| severity       | enum('Low','Medium','High','Critical')    | YES  |     | NULL              |                   |
| cvss_score     | decimal(3,1)                              | YES  |     | NULL              |                   |
| published_date | date                                      | YES  |     | NULL              |                   |
| status         | enum('Open','In Progress','Resolved')     | YES  |     | Open              |                   |
| created_at     | timestamp                                 | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+----------------+-------------------------------------------+------+-----+-------------------+-------------------+
9 rows in set (0.01 sec)
```

Data Query:

```
SELECT * FROM Vulnerabilities;
```

Data Output:

```
mysql> SELECT*FROM VULNERABILITIES;
+---------+----------------+------------+--------------------------------------------+----------+------------+----------------+-------------+---------------------+
| vuln_id | cve_id         | package_id | summary                                    | severity | cvss_score | published_date | status      | created_at          |
+---------+----------------+------------+--------------------------------------------+----------+------------+----------------+-------------+---------------------+
|       1 | CVE-2024-23334 |          1 | Path traversal issue in Werkzeug < 3.0.0   | High     |        8.0 | 2024-05-10     | Open        | 2025-11-13 20:04:14 |
|       2 | CVE-2024-32020 |          3 | Prototype pollution in React DOM           | Critical |        9.6 | 2024-06-09     | Open        | 2025-11-13 20:04:14 |
|       3 | CVE-2023-30861 |          2 | Jinja2 sandbox escape vulnerability        | High     |        8.4 | 2023-09-12     | Resolved    | 2025-11-13 20:04:14 |
|       4 | CVE-2024-29018 |          4 | Express DoS via crafted headers            | High     |        8.0 | 2024-03-19     | In Progress | 2025-11-13 20:04:14 |
|       5 | CVE-2023-48022 |          5 | TensorFlow overflow in quantized kernel    | High     |        8.3 | 2023-12-22     | Open        | 2025-11-13 20:04:14 |
|       6 | CVE-2024-37110 |          6 | NumPy memory leak in reshape()             | Medium   |        6.2 | 2024-02-17     | Open        | 2025-11-13 20:04:14 |
|       7 | CVE-2023-47889 |          9 | Django redirect validation bypass          | Medium   |        5.9 | 2023-11-29     | Resolved    | 2025-11-13 20:04:14 |
|       8 | CVE-2024-23155 |          7 | Pandas CSV parser RCE                      | Critical |        9.8 | 2024-01-19     | Open        | 2025-11-13 20:04:14 |
|       9 | CVE-2024-16002 |          8 | Lodash prototype pollution                 | High     |        8.2 | 2024-01-05     | In Progress | 2025-11-13 20:04:14 |
|      10 | CVE-2024-22221 |         10 | FastAPI header injection vulnerability     | Medium   |        6.0 | 2024-02-12     | Open        | 2025-11-13 20:04:14 |
+---------+----------------+------------+--------------------------------------------+----------+------------+----------------+-------------+---------------------+
10 rows in set (0.05 sec)
```

1.5 `Project_Packages` Table

- **Purpose:** A critical junction table that maps which version of a package is used by which project (the "SBOM" table).

Structure Query:

```
DESCRIBE Project_Packages;
```

Structure Output:

```
mysql> DESC PROJECT_PACKAGES;
+--------------+-------------+------+-----+---------+----------------+
| Field        | Type        | Null | Key | Default | Extra          |
+--------------+-------------+------+-----+---------+----------------+
| projpkg_id   | int         | NO   | PRI | NULL    | auto_increment |
| project_id   | int         | YES  | MUL | NULL    |                |
| package_id   | int         | YES  | MUL | NULL    |                |
| version_used | varchar(50) | YES  |     | NULL    |                |
| last_scanned | date        | YES  |     | NULL    |                |
+--------------+-------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

Data Query:

```
SELECT * FROM Project_Packages;
```

Data Output:

```
mysql> SELECT*FROM PROJECT_PACKAGES;
+------------+------------+------------+--------------+--------------+
| projpkg_id | project_id | package_id | version_used | last_scanned |
+------------+------------+------------+--------------+--------------+
|          1 |          1 |          1 | 2.2.2        | 2025-11-01   |
|          2 |          1 |          2 | 3.0.0        | 2025-11-01   |
|          3 |          2 |          3 | 18.2.0       | 2025-11-03   |
|          4 |          3 |          9 | 5.0.0        | 2025-11-02   |
|          5 |          4 |          5 | 2.15.0       | 2025-10-28   |
|          6 |          5 |          4 | 4.18.0       | 2025-10-30   |
|          7 |          6 |          6 | 1.25.0       | 2025-11-04   |
|          8 |          7 |          7 | 2.1.5        | 2025-11-03   |
|          9 |          8 |          8 | 4.17.20      | 2025-10-29   |
|         10 |         10 |         10 | 0.109.0      | 2025-11-02   |
+------------+------------+------------+--------------+--------------+
10 rows in set (0.05 sec)
```

1.6 `Remediations` Table

- **Purpose:** A junction table that tracks the status of a fix ('Pending', 'Resolved') for a specific vulnerability on a specific project.

Structure Query:

```
DESCRIBE Remediations;
```

Structure Output:

```
mysql> DESC REMEDIATIONS;
+----------------+------------------------------------------------+------+-----+-------------------+-------------------+
| Field          | Type                                           | Null | Key | Default           | Extra             |
+----------------+------------------------------------------------+------+-----+-------------------+-------------------+
| remediation_id | int                                            | NO   | PRI | NULL              | auto_increment    |
| project_id     | int                                            | YES  | MUL | NULL              |                   |
| vuln_id        | int                                            | YES  | MUL | NULL              |                   |
| fixed_version  | varchar(50)                                    | YES  |     | NULL              |                   |
| status         | enum('Pending','In Progress','Resolved')       | YES  |     | Pending           |                   |
| date_applied   | date                                           | YES  |     | NULL              |                   |
| created_at     | timestamp                                      | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+----------------+------------------------------------------------+------+-----+-------------------+-------------------+
7 rows in set (0.00 sec)
```

Data Query:

```
SELECT * FROM Remediations;
```

Data Output:

```
mysql> SELECT*FROM REMEDIATIONS;
+----------------+------------+---------+---------------+-------------+--------------+---------------------+
| remediation_id | project_id | vuln_id | fixed_version | status      | date_applied | created_at          |
+----------------+------------+---------+---------------+-------------+--------------+---------------------+
|             11 |          1 |       2 | 3.0.1         | In Progress | 2025-11-13   | 2025-11-13 21:12:15 |
|             12 |          1 |       2 | 3.0.1         | In Progress | 2025-11-13   | 2025-11-13 21:12:41 |
+----------------+------------+---------+---------------+-------------+--------------+---------------------+
2 rows in set (0.04 sec)
```

1.7 `Audit_Log` Table

- **Purpose:** A log table that records user actions for accountability.

Structure Query:

```
DESCRIBE Audit_Log;
```

Structure Output:

```
mysql> DESC AUDIT_LOG;
+-----------+--------------+------+-----+-------------------+-------------------+
| Field     | Type         | Null | Key | Default           | Extra             |
+-----------+--------------+------+-----+-------------------+-------------------+
| log_id    | int          | NO   | PRI | NULL              | auto_increment    |
| user_id   | int          | YES  | MUL | NULL              |                   |
| action    | varchar(255) | YES  |     | NULL              |                   |
| details   | text         | YES  |     | NULL              |                   |
| timestamp | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----------+--------------+------+-----+-------------------+-------------------+
5 rows in set (0.00 sec)
```

Data Query:

```
SELECT * FROM Audit_Log;
```

Data Output:

```
mysql> SELECT*FROM AUDIT_LOG;
+--------+---------+-----------------------+-------------------------------------------------+---------------------+
| log_id | user_id | action                | details                                         | timestamp           |
+--------+---------+-----------------------+-------------------------------------------------+---------------------+
|      1 |       1 | Added vulnerability   | Inserted CVE-2024-23334 into database           | 2025-11-13 20:04:32 |
|      2 |       2 | Scanned project       | Scanned Flask for outdated packages             | 2025-11-13 20:04:32 |
|      3 |       3 | Updated remediation   | Marked CVE-2023-30861 as Resolved               | 2025-11-13 20:04:32 |
|      4 |       4 | Added package         | Added React version mapping                     | 2025-11-13 20:04:32 |
|      5 |       5 | Updated vulnerability | Changed status of CVE-2024-29018 to In Progress | 2025-11-13 20:04:32 |
|      6 |       6 | System Scan           | Performed full ecosystem vulnerability scan     | 2025-11-13 20:04:32 |
|      7 |       7 | Resolved issue        | Patched Lodash in Angular project               | 2025-11-13 20:04:32 |
|      8 |       8 | Added project         | Added TensorFlow to VMDB_OSS                    | 2025-11-13 20:04:32 |
|      9 |       9 | Updated project       | Modified React repo link                        | 2025-11-13 20:04:32 |
|     10 |      10 | Added new CVE         | Inserted CVE-2024-22221 for FastAPI             | 2025-11-13 20:04:32 |
+--------+---------+-----------------------+-------------------------------------------------+---------------------+
10 rows in set (0.03 sec)
```

2.0 Views (Analytical Layer)

This section details the 6 views that provide pre-joined, human-readable analytics.

2.1 `vw_all_vulnerabilities` View

- **Purpose:** Lists all known vulnerabilities with their human-readable package names.

Structure Query:

```
SHOW CREATE VIEW vw_all_vulnerabilities;
```

Data Query:

```
SELECT * FROM vw_all_vulnerabilities;
```

Data Output:

```
mysql> SELECT * FROM vw_all_vulnerabilities;
+---------+----------------+--------------+----------+-------------+----------------+
| vuln_id | cve_id         | package_name | severity | status      | published_date |
+---------+----------------+--------------+----------+-------------+----------------+
|       1 | CVE-2024-23334 | Werkzeug     | High     | Open        | 2024-05-10     |
|       2 | CVE-2024-32020 | React        | Critical | Open        | 2024-06-09     |
|       3 | CVE-2023-30861 | Jinja2       | High     | Resolved    | 2023-09-12     |
|       4 | CVE-2024-29018 | Express      | High     | In Progress | 2024-03-19     |
|       5 | CVE-2023-48022 | TensorFlow   | High     | Open        | 2023-12-22     |
|       6 | CVE-2024-37110 | NumPy        | Medium   | Open        | 2024-02-17     |
|       7 | CVE-2023-47889 | Django       | Medium   | Resolved    | 2023-11-29     |
|       8 | CVE-2024-23155 | Pandas       | Critical | Open        | 2024-01-19     |
|       9 | CVE-2024-16002 | Lodash       | High     | In Progress | 2024-01-05     |
|      10 | CVE-2024-22221 | FastAPI      | Medium   | Open        | 2024-02-12     |
+---------+----------------+--------------+----------+-------------+----------------+
10 rows in set (0.00 sec)
```

2.2 `vw_project_vulnerabilities` View

- **Purpose:** The main risk-register view. Shows which projects are affected by which vulnerabilities and the remediation status.

Structure Query:

```
SHOW CREATE VIEW vw_project_vulnerabilities;
```

Data Query:

```
SELECT * FROM vw_project_vulnerabilities;
```

Data Output:

```
mysql> SELECT * FROM vw_project_vulnerabilities;
+--------------+--------------+----------------+----------+--------------------+
| project_name | package_name | cve_id         | severity | remediation_status |
+--------------+--------------+----------------+----------+--------------------+
| Flask        | Werkzeug     | CVE-2024-23334 | High     | NULL               |
| Flask        | Jinja2       | CVE-2023-30861 | High     | NULL               |
| React        | React        | CVE-2024-32020 | Critical | NULL               |
| Django       | Django       | CVE-2023-47889 | Medium   | NULL               |
| TensorFlow   | TensorFlow   | CVE-2023-48022 | High     | NULL               |
| ExpressJS    | Express      | CVE-2024-29018 | High     | NULL               |
| NumPy        | NumPy        | CVE-2024-37110 | Medium   | NULL               |
| Pandas       | Pandas       | CVE-2024-23155 | Critical | NULL               |
| Angular      | Lodash       | CVE-2024-16002 | High     | NULL               |
| FastAPI      | FastAPI      | CVE-2024-22221 | Medium   | NULL               |
+--------------+--------------+----------------+----------+--------------------+
10 rows in set (0.02 sec)
```

2.3 `vw_project_package_usage` View

- **Purpose:** The "Software Bill of Materials" view. Shows which projects use which packages and at what version.

Structure Query:

```
SHOW CREATE VIEW vw_project_package_usage;
```

Data Query:

```
SELECT * FROM vw_project_package_usage;
```

Data Output:

```
mysql> SELECT * FROM vw_project_package_usage;
+--------------+--------------+--------------+--------------+
| project_name | package_name | version_used | last_scanned |
+--------------+--------------+--------------+--------------+
| Flask        | Werkzeug     | 2.2.2        | 2025-11-01   |
| Flask        | Jinja2       | 3.0.0        | 2025-11-01   |
| React        | React        | 18.2.0       | 2025-11-03   |
| Django       | Django       | 5.0.0        | 2025-11-02   |
| TensorFlow   | TensorFlow   | 2.15.0       | 2025-10-28   |
| ExpressJS    | Express      | 4.18.0       | 2025-10-30   |
| NumPy        | NumPy        | 1.25.0       | 2025-11-04   |
| Pandas       | Pandas       | 2.1.5        | 2025-11-03   |
| Angular      | Lodash       | 4.17.20      | 2025-10-29   |
| FastAPI      | FastAPI      | 0.109.0      | 2025-11-02   |
+--------------+--------------+--------------+--------------+
10 rows in set (0.00 sec)
```

2.4 `vw_package_vuln_count` View

- **Purpose:** An aggregate view that counts how many vulnerabilities are associated with each package.

Structure Query:

```
SHOW CREATE VIEW vw_package_vuln_count;
```

Data Query:

```
SELECT * FROM vw_package_vuln_count;
```

Data Output:

```
mysql> SELECT * FROM vw_package_vuln_count;
+--------------+---------------------+
| package_name | vulnerability_count |
+--------------+---------------------+
| Werkzeug     |                   1 |
| Jinja2       |                   1 |
| React        |                   1 |
| Express      |                   1 |
| TensorFlow   |                   1 |
| NumPy        |                   1 |
| Pandas       |                   1 |
| Lodash       |                   1 |
| Django       |                   1 |
| FastAPI      |                   1 |
+--------------+---------------------+
10 rows in set (0.02 sec)
```

2.5 `vw_remediation_progress` View

- **Purpose:** A "dashboard widget" view. Shows a high-level count of all remediations by status.

Structure Query:

```
SHOW CREATE VIEW vw_remediation_progress;
```

Data Query:

```
SELECT * FROM vw_remediation_progress;
```

Data Output:

```
mysql> SELECT * FROM vw_remediation_progress;
+----------------+------------------+---------------+
| resolved_total | inprogress_total | pending_total |
+----------------+------------------+---------------+
|              0 |                2 |             0 |
+----------------+------------------+---------------+
1 row in set (0.02 sec)
```

2.6 `vw_critical_vulnerabilities` View

- **Purpose:** A high-priority filtered view that shows *only* vulnerabilities marked as 'Critical'.

Structure Query:

```
SHOW CREATE VIEW vw_critical_vulnerabilities;
```

Data Query:

```
SELECT * FROM vw_critical_vulnerabilities;
```

Data Output:

```
mysql> SELECT * FROM vw_critical_vulnerabilities;
+----------------+--------------+-----------------------------------+----------+--------+
| cve_id         | package_name | summary                           | severity | status |
+----------------+--------------+-----------------------------------+----------+--------+
| CVE-2024-32020 | React        | Prototype pollution in React DOM  | Critical | Open   |
| CVE-2024-23155 | Pandas       | Pandas CSV parser RCE             | Critical | Open   |
+----------------+--------------+-----------------------------------+----------+--------+
2 rows in set (0.02 sec)
```

3.0 Stored Procedures & Functions

This section details the 3 routines that provide a safe API for database operations.

3.1 `AddVulnerabilitySimple` Procedure

- **Purpose:** Safely inserts a new vulnerability into the `Vulnerabilities` table.

Structure Query:

```
SHOW CREATE PROCEDURE AddVulnerabilitySimple;
```

Execution & Verification Query:

```
-- This call will add a new test vulnerability
CALL AddVulnerabilitySimple('CVE-2025-9999', 1, 'Test vulnerability via procedure',
'Medium', 5.0);

-- This query verifies the procedure worked
SELECT * FROM Vulnerabilities WHERE cve_id = 'CVE-2025-9999';
```

3.2 `UpdateVulnerabilityStatus` Procedure

- **Purpose:** Safely updates the *global* status of an existing vulnerability.

Structure Query:

```
SHOW CREATE PROCEDURE UpdateVulnerabilityStatus;
```

Execution & Verification Query:

```
-- This call will change the status of vuln_id 1
CALL UpdateVulnerabilityStatus(1, 'In Progress');
```

```
-- This query verifies the procedure worked
SELECT * FROM Vulnerabilities WHERE vuln_id = 1;
```

Data Output:

```
mysql> -- This call will change the status of vuln_id 1
mysql> CALL UpdateVulnerabilityStatus(1, 'In Progress');
Query OK, 1 row affected (0.04 sec)

mysql>
mysql> -- This query verifies the procedure worked
mysql> SELECT * FROM Vulnerabilities WHERE vuln_id = 1;
+---------+----------------+------------+-------------------------------------------+----------+------------+----------------+-------------+---------------------+
| vuln_id | cve_id         | package_id | summary                                   | severity | cvss_score | published_date | status      | created_at          |
+---------+----------------+------------+-------------------------------------------+----------+------------+----------------+-------------+---------------------+
|       1 | CVE-2024-23334 |          1 | Path traversal issue in Werkzeug < 3.0.0 | High     |        8.0 | 2024-05-10     | In Progress | 2025-11-13 20:04:14 |
+---------+----------------+------------+-------------------------------------------+----------+------------+----------------+-------------+---------------------+
1 row in set (0.00 sec)
```

3.3 `CountVulnsForPackage` Function

- **Purpose:** A utility function that returns an integer count of vulnerabilities for a given package ID.

Structure Query:

```
SHOW CREATE FUNCTION CountVulnsForPackage;
```

Execution Query:

```
-- This query uses the function to count vulns for package_id 1 ('Werkzeug')
SELECT CountVulnsForPackage(1) AS WerkzeugVulnCount;
```